Fitting item response theory models with a nonnormal latent trait

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Item response theory

- Item response theory (IRT) is a widely used tool to link responses on a set of items from a test, inventory, or survey to the latent trait that the instrument is designed to measure.

- There exist a number of models for estimating this relationship.
  - Models differ in terms of the item parameters estimated and the type of item response.

- Researchers are often interested in obtaining an estimate of the trait for individual respondents as well as qualities of the items; e.g., difficulty and discrimination.

IRT models

- Common IRT models include the Rasch, 1-parameter logistic (1PL), 2-parameter logistic (2PL) and 3-parameter logistic (3PL) for dichotomous items and the graded response model (GRM) for polytomous items.

- 3PL model:

  $$P(x_j = 1|\theta, b_j) = c_j + (1 - c_j)\frac{e^{a_j(\theta, -b_j)}}{1+e^{a_j(\theta, -b_j)}}$$

  Where
  θ =Latent trait score
  a_j =Discrimination parameter for item *j*
  b_j =Difficulty parameter for item *j*
  c_j =Pseudo-chance parameter value for item *j*

- The 2PL model removes terms associated with c_j.
- The 1PL model constrains item discrimination to be equal across items.
- The Rasch model constrains item discrimination to be 1 across items.

IRT parameter estimation

- There are two common ways to estimate item and person parameters for IRT models:

1. Joint maximum likelihood (JML) – Makes no assumptions about latent trait distribution but is highly parameterized.

2. Marginal maximum likelihood (MML) – More efficient to estimate than JML, but rests on the latent trait normality assumption.

- MML is more widely used than JML because it is more efficient and computationally less demanding.

IRT estimation and a nonnormal latent trait

- MML estimation is done using the expectation-maximization (EM) algorithm and rests on an assumption of normality of the latent trait.

- This assumption may not be reasonable in many situations.
  - Skewed distribution for traits such as anxiety or depression.
  - Bimodal distribution for cognitive skills with higher and lower achieving groups.
  - Zero-inflated for relatively rare conditions where many members of the population have no level of the trait; e.g. no symptoms of schizophrenia.

Impact of nonnormality on IRT parameter estimation

- When the latent trait is not normally distributed MML can yield biased estimates for both item and person parameters.

- Research has shown that for any symmetric distribution there is relatively little bias in model parameter estimates (Li & Lee, 2026).

- When the distribution is skewed or zero-inflated, however, parameter bias can be very pronounced, particularly for more extreme items (Woods, 2015).

Methods for IRT parameter estimation with nonnormal latent traits

- Several methods for IRT parameter estimation in the presence of nonnormal latent traits have been developed.

- These techniques have in common that they attempt to account for the actual shape of the latent trait distribution rather than working under the assumption that it is normal.

- These methods include the following:
  - Empirical histogram method
  - Log-linear smoothing
  - Davidian curves
  - Kernel smoothing
  - Zero-inflated IRT
  - Bayesian estimation

Empirical histogram method

- The empirical histogram method (EHM) estimates the latent trait distribution, $g(\theta)$, by fitting a histogram at each quadrature point.
  - The histogram height at each point is calculated as the sum of posterior probabilities of respondents having a score at that value.

- The histogram reflects the number of individuals with a score at a given quadrature point.

- The shape of the histogram is updated at each step of the EM algorithm and item parameter values (difficulty and discrimination) are estimated based on this latent trait estimate.

Empirical histogram method

- EHM can better reflect the actual shape of the latent trait distribution rather than assuming it is normally distributed.

- EHM does have some potential problems, however.
  1. Sensitive to the range and number of quadrature points.
  2. Can overfit the sample data and lead to a less generalizable model.
  3. The latent trait curve is not smooth.
  4. The number of parameters to be estimated for EHM can be quite large.
     - The number of free parameters estimated for EHM is equal to the number of quadrature points minus 3; e.g., for 70 quadrature points there would be 67 free parameters to estimate.
     - However, having more quadrature points yields a latent trait distribution estimate that is potentially a better representation of the actual distribution.

Log-linear smoothing

- One approach to addressing the noise and overfitting that can come with EHM is to smooth the latent trait estimates using a log-linear model (Casabianca & Lewis, 2015).
- This approach, known as log-linear smoothing (LLS), fits a log-linear model to the estimated frequencies produced by EHM.
- These frequencies are modeled as a combination of multiple polynomial functions.
- The highest polynomial degree (complexity of the function) is set by the researcher.
    - There is not an established approach for determining how to identify the optimal polynomial degree to use.

Davidian curves

- The principle underlying EHM is to estimate the actual shape of the latent trait rather than assuming a particular functional form.
- An alternative approach is to estimate the trait density using complex mathematical functions, such as splines.
- Davidian curves (DCs) use a combination of the normal density and a smoothing function (Woods, 2009) to estimate the latent trait.

Davidian curves

- DC is a semi-parametric approach that combines the normal distribution for the latent trait with a smoothing parameter to account for nonnormality in the empirical distribution.

- The latent trait is first estimated using DC and then used to estimate the item parameters.

- In this way, nonnormality of the trait can be incorporated into estimation of the other parameters, thereby mitigating the bias imbued when assuming a normally distributed latent trait.

Davidian curves

- With DC, the latent trait ($h(\theta)$) takes the form:

$h(\theta) = R_t^2(\theta)g(\theta)$
where
$g(\theta) =$Normal density
$t =$Number of smoothing parameters.
$R_t(\theta) = \sum_{\lambda}^{t} m_\lambda \theta^\lambda$
$\lambda =$Nonnegative integer
$m =$Vector of smoothing parameters where $m \neq 0$

- R_t is a polynomial of degree t where higher values make the density more flexible.
    - Greater flexibility allows for a more complex distribution of the latent trait.

Davidian curves

- There are $t + 1$ coefficients in the smoothing function.

- As an example, if $t = 1$, the smoothing function takes the form:
  - $R_k^2(\theta) = (m_0 + m_1\theta)^2$

- In the estimation process, several values of t are fit to the data and the model with the smallest Hannan-Quin information criterion (HQIC) is selected.

- Research (e.g., Woods, 2015) has shown that $t = 1$ or 2 typically accommodates the presence of multimodality or skewness.

Kernel density estimation of latent trait

- The advantage of DC in IRT parameter estimation is that it allows for nonnormality of the latent trait and produces a smooth function, unlike EHM.

- Another smoothing based approach for estimating the latent trait is kernel density estimation (KDE).

- KDE is a widely used technique throughout statistics.

- Li and Lee (2026) extended its use to the case of a nonnormal latent trait in IRT.

KDE for IRT

- KDE doesn't make any assumptions about the underlying shape of the distribution in question.
  - This is in contrast to DC, which is semiparametric and assumes an underlying normality that is then smoothed.

- KDE estimates the value for a given point in the distribution by using only neighboring points, rather than the full sample.
  - In this way, it can capture the correct distributional shape by accounting for localized anomalies.

- The size of this neighborhood is determined by the bandwidth, h .

KDE for IRT

- The kernel function formulates the latent trait density to be:

$$g(z|h) = \frac{1}{Nh}\sum_{j=1}^{N} K\left(\frac{z-Z_j}{h}\right)$$

Where

h =Bandwidth

$K(\cdot)$ =Kernel function

Z_j =Observed data points

N =Sample size

KDE for IRT

- The Gaussian kernel is the most common technique because it has the least influence on the resulting density estimates.

- When the Gaussian kernel is used, h is the standard deviation of the kernels.

- The value of h determines the degree of smoothing.
  - Similar to *t* in DCs.

- The optimal h can be selected automatically using the asymptotic mean integrated squared error (AMISE).
  - Obviates the need to fit many models and manually select the best one, as with DC.

Zero-inflated mixture IRT model

- In some situations, the construct being measured may not be present for a large portion of the population and normally distributed for the remainder.

- The result of this situation is that the latent trait may be bifurcated with one portion being represented by a spike at the far left side of the distribution.

- This distributional shape conforms to the zero-inflated (ZI) IRT model.

Zero-inflated mixture IRT model

- For estimation of the ZI model, the normally distributed latent trait is replaced with a mixture of two distributions.
  - One of these distributions is centered at a very small latent trait value (-100).
  - The other distribution is the standard normal.

- Formally, this model is:

  $f(\theta) = \pi I(\theta = -100) + (1 - \pi) g(\theta)$

  Where

  $g(\theta) =$Standard normal distribution

  $\pi =$Probability that the latent trait is in the ZI condition for a randomly selected individual.

- This model is only appropriate when the latent trait distribution is ZI.

Bayesian estimator

- The final approach for estimating the IRT model parameters that we will consider is the Bayesian estimator.

- Informally, we can conceptualize Bayesian statistics as involving two pieces of information germane to our understanding the population parameter: The data and a prior distribution placed on the population parameter (e.g. the prior for the latent trait could be the standard normal distribution).

- The likelihood and the prior are then combined to form the distribution of the parameter of interest (e.g. latent trait) in what is typically referred to as the posterior distribution.

Bayesian framework for IRT models

- We can express item response data in the Bayesian framework using the following equality.
  - $f(\Omega|X) = f(X|\Omega) * f(\Omega) / \left[\int_{\Omega} (f(X|\Omega) * f(\Omega) d\Omega)\right]$
  - Where
  - X =All item responses for all examinees
  - Ω =Unknown parameters
- $f(\Omega|X)$ is the posterior probability of a particular set of IRT model parameters given the current data.
- $f(\Omega)$ is the prior probability of the model parameters.
- $f(X|\Omega)$ is the probability of getting the item responses that we have, given a particular set of model parameters; i.e. this is the maximum likelihood estimate of the item parameters.
- The denominator represents the overall probability of our obtaining the data that we have, across all possible values of the model parameters.

Bayesian framework for IRT models

- $f(\Omega|X)$, the posterior distribution of the model parameters, is our estimate of the actual distribution of the parameters of interest (i.e. item difficulty, discrimination, pseudo-chance, and person ability).

- Typically $f(\Omega|X)$ is too complex to estimate analytically.

- It is, however, often possible to sample from $f(\Omega|X)$ to create an estimate, which we refer to as the posterior distribution.

- MCMC is the mechanism by which we sample from the distribution of interest in order to create the posterior distribution.

Bayesian framework for IRT models

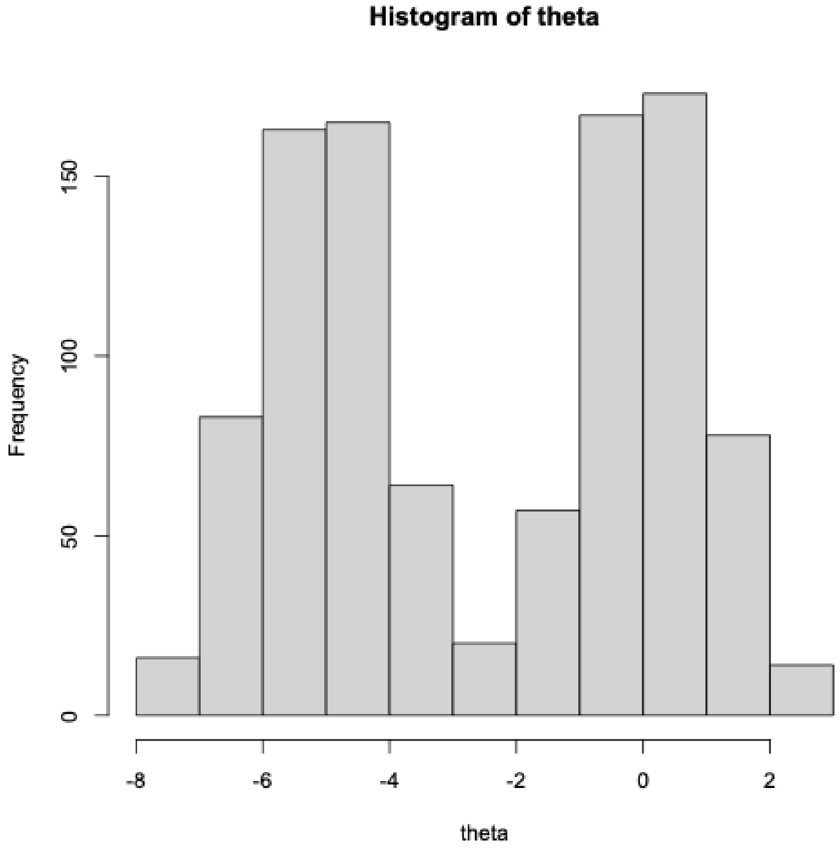
- The prior distribution of the latent trait is typically the normal with mean of 0 and a large standard deviation (e.g., 10).
- Typical priors for item difficulty are the normal (0,10) and for item discrimination the lognormal (0, 10).
- Bayesian estimation has been shown to be more robust to departures from normality of the latent trait (e.g., Baker & Kim, 20004; Ansari & Jedidi, 2000).

Example

- We will now apply each of these methods to two example datasets.
- The first dataset consist of simulated dichotomous item responses that were generated with a bimodal latent trait
- The second dataset also included a simulated set of dichotomous items, in this case with a skewed latent.
- We'll fit the data with each of the techniques that are described above, as well as the standard 2PL IRT model.

Bimodal latent trait: Data generation

- Sample size is 1000.
- 20 dichotomous items.
- Bimodal latent trait.
- Item discrimination values range between 0.8 and 1.5.
- Item difficulties range between -2 and 2.



Bimodal latent trait: Parameter estimates

Item Discrimination

	Population	MML	KDE	EHM	LLS	Davidian5	Bayes
Item_1	1.0	2.249633	2.571879	1.1181582	2.702756	2.987220	1.862804
Item_2	1.0	2.007721	2.186468	0.9501579	2.215459	2.550324	1.870239
Item_3	1.0	2.556100	3.039195	1.3069974	3.273937	3.546255	1.959524
Item_4	1.0	2.225361	2.663896	1.2142185	2.907392	2.942593	1.860868
Item_5	1.5	3.126662	3.713395	1.6051336	4.073289	4.399563	1.947319
Item_6	1.5	3.289524	3.995109	1.7331806	4.543672	4.696709	1.942536
Item_7	1.5	3.750502	4.451450	1.7546609	4.913324	5.315095	1.930200
Item_8	1.5	3.439682	4.148727	1.7174692	4.661035	4.900880	1.914158
Item_9	0.8	2.613407	2.531330	1.0711647	2.431242	2.993434	1.967819
Item_10	0.8	2.156731	2.108817	0.8824549	2.041395	2.497058	1.938383
Item_11	0.8	2.232782	2.099774	0.9042542	2.006644	2.481187	1.960298
Item_12	0.8	2.340633	2.290995	0.9579941	2.218664	2.723313	1.942322
Item_13	1.0	3.374031	3.229309	1.3115115	3.091654	3.812972	1.999920
Item_14	1.0	2.777381	2.544810	1.0776476	2.413400	3.007792	2.060578
Item_15	1.0	2.857345	2.648124	1.1147405	2.513376	3.115306	2.055215
Item_16	1.0	2.867663	2.539633	1.0916122	2.370277	2.989265	1.990701
Item_17	1.5	5.883015	5.097198	2.1236032	4.706763	5.844097	2.039691
Item_18	1.5	4.329234	3.897029	1.6138395	3.648747	4.548769	2.050148
Item_19	1.5	4.546507	4.012505	1.6962056	3.743580	4.629460	2.060796
Item_20	1.5	4.717181	4.151679	1.7655614	3.898804	4.758669	2.097377

Item Difficulty

	Population	MML	KDE	EHM	LLS	Davidian5	Bayes
Item_1	-2	-3.829336	1.6413307	2.0335614	1.6218763	1.4065841	-0.9276326
Item_2	-2	-3.687470	1.7902585	2.3728306	1.7883691	1.5334230	-0.9980451
Item_3	-2	-4.261979	1.5934763	1.9367292	1.5651824	1.3624001	-0.9402392
Item_4	-2	-3.886473	1.6516363	1.9968095	1.6115031	1.4415571	-0.9757706
Item_5	-1	-3.693749	1.1975429	1.0434188	1.2020100	1.0194590	-0.4360868
Item_6	-1	-3.859198	1.1889310	1.0276295	1.1906664	1.0130272	-0.4384423
Item_7	-1	-4.018788	1.1117635	0.8740195	1.1263537	0.9459957	-0.4449796
Item_8	-1	-4.104103	1.2077386	1.0869921	1.2092682	1.0281880	-0.4128594
Item_9	0	-2.162099	0.8881471	0.3011418	0.9092797	0.7588518	0.0392698
Item_10	0	-2.064608	1.0086190	0.5919291	1.0291062	0.8611433	0.0749265
Item_11	0	-1.961276	0.9369142	0.4094194	0.9577157	0.8009254	0.0378232
Item_12	0	-1.992590	0.9053352	0.3454392	0.9245022	0.7731366	0.0662695
Item_13	1	-1.871948	0.6231941	-0.3480080	0.6409944	0.5344032	0.5245512
Item_14	1	-1.355718	0.5331809	-0.5424298	0.5396544	0.4584646	0.6139325
Item_15	1	-1.549333	0.5950959	-0.3985614	0.6057662	0.5105213	0.5721966
Item_16	1	-1.500075	0.5720402	-0.4497042	0.5795531	0.4912394	0.5871874
Item_17	2	-2.144810	0.4437050	-0.8326239	0.4562899	0.3803245	0.9854540
Item_18	2	-1.674477	0.4498672	-0.7794807	0.4593220	0.3854454	1.0294736
Item_19	2	-1.842722	0.4722120	-0.7262231	0.4833090	0.4033700	1.0478972
Item_20	2	-1.907702	0.4737941	-0.7257828	0.4870695	0.4040649	1.0288951

Bimodal latent trait: Model fit

```
The best model: results.kde

                           HQ
results.mml2         13054.04
results.kde          12788.62
results.lls          12801.20
results.ehm          13406.85
results.davidian5 12850.18
```

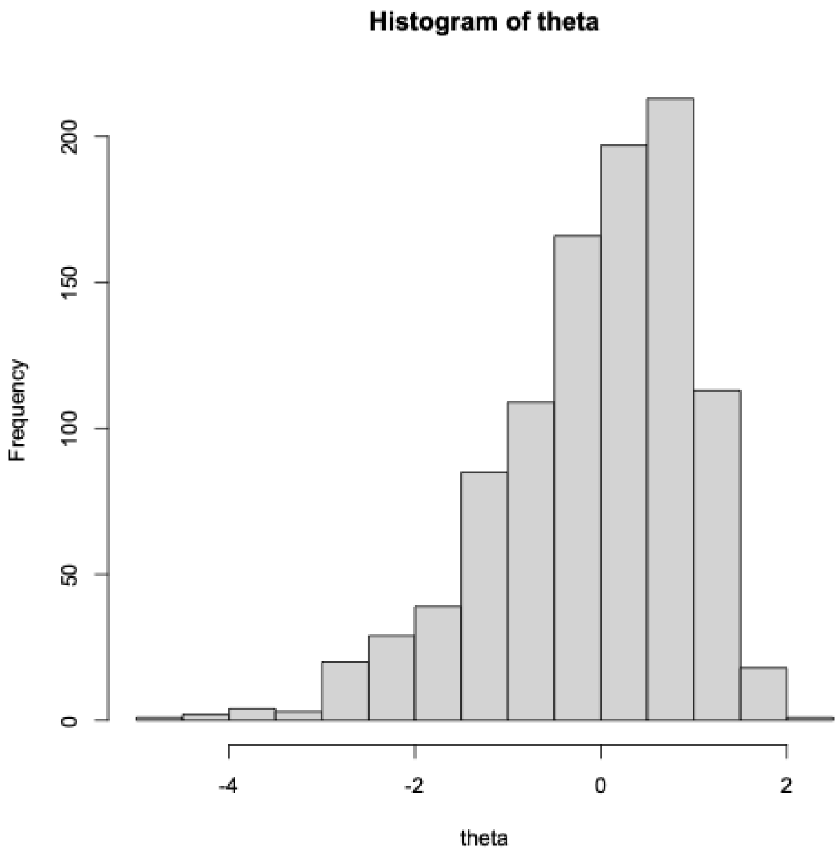
Estimator	AIC	BIC
MML	13417.39	13613.70
KDE	13216.07	13417.29
LLS	13225.04	13440.99
EHM	13578.78	14359.12
Davidian5	13252.40	13473.25

Bimodal latent trait: Model fit

```
Result of model comparison

              logLik deviance      AIC      BIC       HQ n_pars     chi p_value
results.mml2 -6668.694 13337.39 13417.39 13613.70 13492.00     40      NA      NA
results.kde  -6567.036 13134.07 13216.07 13417.29 13292.55     41 203.316       0

Result of model comparison

              logLik deviance      AIC      BIC       HQ n_pars      chi p_value
results.mml2 -6668.694 13337.39 13417.39 13613.70 13492.00     40       NA      NA
results.lls  -6568.522 13137.04 13225.04 13440.99 13307.12     44 50.08574       0

Result of model comparison

              logLik deviance      AIC      BIC       HQ n_pars       chi p_value
results.mml2 -6668.694 13337.39 13417.39 13613.70 13492.00     40        NA      NA
results.ehm  -6630.392 13260.78 13578.78 14359.12 13875.36    159 0.6437323       1

Result of model comparison

                    logLik deviance      AIC      BIC       HQ n_pars     chi p_value
results.mml2      -6668.694 13337.39 13417.39 13613.70 13492.00     40      NA      NA
results.davidian5 -6581.201 13162.40 13252.40 13473.25 13336.34     45 34.9972       0
```

Skewed latent trait: Data generation

- Sample size is 1000.
- 20 dichotomous items.
- Negatively skewed latent trait.
- Item discrimination values range between 0.8 and 1.5.
- Item difficulties range between -2 and 2.



Skewed latent trait: Parameter estimates

Item Discrimination

	Population	MML	KDE	EHM	LLS	Davidian5	Bayes
Item_1	1.0	0.9401757	1.0379388	1.1278237	1.0995876	1.0663239	0.9405831
Item_2	1.0	0.7446453	0.8227887	0.8756013	0.8739793	0.8574041	0.8009462
Item_3	1.0	1.0281231	1.1139421	1.1805263	1.1697619	1.1558026	1.0083625
Item_4	1.0	0.8460739	0.9311262	0.9930469	0.9867905	0.9659443	0.8683006
Item_5	1.5	1.2244379	1.2969002	1.3620419	1.3485733	1.3354926	1.1864688
Item_6	1.5	1.5286475	1.6492091	1.7523033	1.7396346	1.7246725	1.4457558
Item_7	1.5	1.3511878	1.4746615	1.5878484	1.5627953	1.5375579	1.2984850
Item_8	1.5	1.3209071	1.4316297	1.5242103	1.5106838	1.4930374	1.2761342
Item_9	0.8	0.8272401	0.8236972	0.8301942	0.8300755	0.8335886	0.8594655
Item_10	0.8	0.8288464	0.8369880	0.8543342	0.8503607	0.8514464	0.8598807
Item_11	0.8	0.8277501	0.8377028	0.8538828	0.8508424	0.8517936	0.8579438
Item_12	0.8	0.7797160	0.7652698	0.7671356	0.7664115	0.7723584	0.8185835
Item_13	1.0	1.2437835	1.1901624	1.1820314	1.1831655	1.1950346	1.2409320
Item_14	1.0	0.9828875	0.9486849	0.9462087	0.9474038	0.9543413	1.0109417
Item_15	1.0	1.1584295	1.1202105	1.1202409	1.1179229	1.1257268	1.1579673
Item_16	1.0	1.0516111	1.0060457	1.0004860	0.9994164	1.0092247	1.0732900
Item_17	1.5	1.9706627	1.7796202	1.7539161	1.7327544	1.7530020	1.8610562
Item_18	1.5	1.6008331	1.4477802	1.4178405	1.4108567	1.4302463	1.5506274
Item_19	1.5	1.8508539	1.6806188	1.6470431	1.6379983	1.6575358	1.7694689
Item_20	1.5	1.6359825	1.4851802	1.4603897	1.4479804	1.4637605	1.5823139

Item Difficulty

	Population	MML	KDE	EHM	LLS	Davidian5	Bayes
Item_1	-2	-1.9749403	1.9498109	1.8248628	1.8597542	1.8979521	-2.1044547
Item_2	-2	-2.1370690	2.6389567	2.4964669	2.5025830	2.5391081	-2.7101486
Item_3	-2	-2.0572992	1.8858227	1.7971324	1.8106701	1.8229027	-2.0362407
Item_4	-2	-1.9107248	2.0951466	1.9843128	1.9955115	2.0245698	-2.2186221
Item_5	-1	-0.9707477	0.7898048	0.7750596	0.7765534	0.7718546	-0.8086575
Item_6	-1	-1.0266194	0.6782543	0.6715332	0.6697457	0.6614081	-0.6894421
Item_7	-1	-1.1338046	0.8242578	0.8014992	0.8042869	0.8011406	-0.8589464
Item_8	-1	-1.0206740	0.7649026	0.7490119	0.7491975	0.7439554	-0.7897296
Item_9	0	0.0038813	0.0139376	0.0219656	0.0196025	0.0130568	-0.0021620
Item_10	0	-0.0651725	0.0975035	0.1049772	0.1026067	0.0956564	-0.0812534
Item_11	0	-0.0559232	0.0864943	0.0941797	0.0917781	0.0848358	-0.0713797
Item_12	0	0.0368920	-0.0316563	-0.0246564	-0.0269652	-0.0326796	0.0392986
Item_13	1	1.1169955	-0.9051541	-0.8954757	-0.8987925	-0.8985175	0.8995409
Item_14	1	0.9418825	-0.9690263	-0.9601267	-0.9624798	-0.9626552	0.9375111
Item_15	1	0.8609421	-0.7388749	-0.7250821	-0.7299199	-0.7327707	0.7437714
Item_16	1	0.9256298	-0.8918730	-0.8838965	-0.8881266	-0.8873635	0.8669338
Item_17	2	2.1501226	-1.1329929	-1.1258437	-1.1387863	-1.1360496	1.1211453
Item_18	2	1.8752477	-1.2299096	-1.2316924	-1.2406859	-1.2345983	1.1935045
Item_19	2	2.1914449	-1.2357008	-1.2347060	-1.2446147	-1.2402954	1.2115631
Item_20	2	2.0643304	-1.3265262	-1.3267794	-1.3391588	-1.3340303	1.2875824

Skewed latent trait: Model fit

```
The best model: results.kde

                              HQ
results.mml2        21049.99
results.kde         21021.94
results.lls         21029.72
results.ehm         21473.53
results.davidian5 21034.38
```

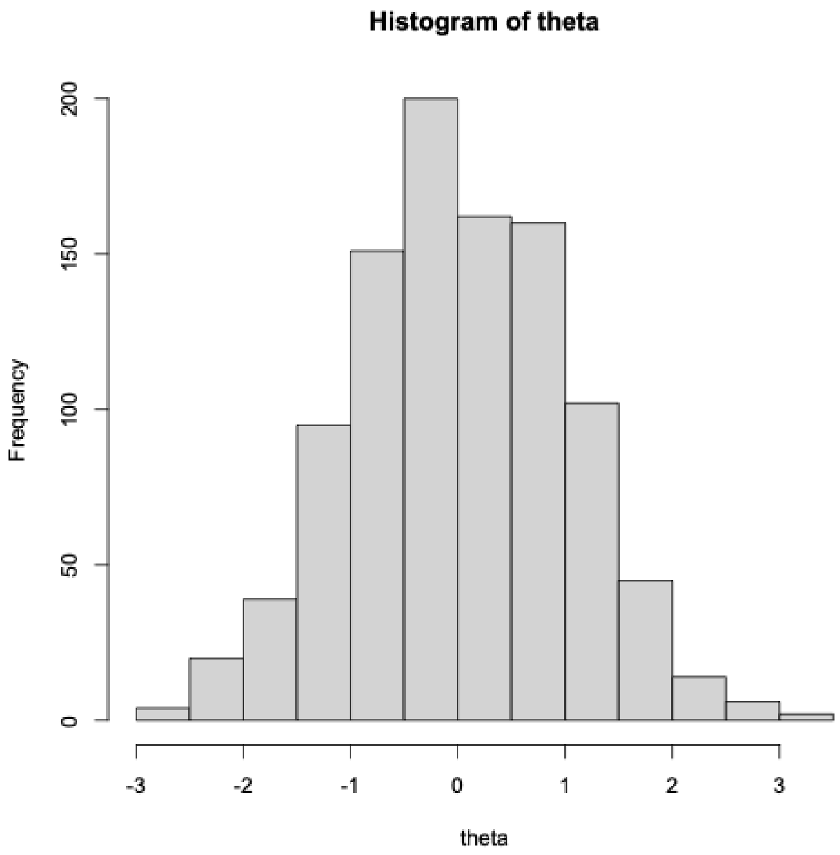
Estimator	AIC	BIC
MML	20975.38	21171.69
KDE	20945.46	21146.68
LLS	20947.65	21163.59
EHM	21176.94	21957.28
Davidian5	20950.45	21171.29

Skewed latent trait: Model fit

```
Result of model comparison

               logLik deviance      AIC      BIC       HQ n_pars      chi p_value
results.mml2 -10447.69 20895.38 20975.38 21171.69 21049.99     40       NA      NA
results.kde  -10431.73 20863.46 20945.46 21146.68 21021.94     41 31.91924       0

Result of model comparison

               logLik deviance      AIC      BIC       HQ n_pars      chi p_value
results.mml2 -10447.69 20895.38 20975.38 21171.69 21049.99     40       NA      NA
results.lls  -10429.82 20859.65 20947.65 21163.59 21029.72     44 8.933742  0.0628

Result of model comparison

               logLik deviance      AIC      BIC       HQ n_pars      chi p_value
results.mml2 -10447.69 20895.38 20975.38 21171.69 21049.99     40       NA      NA
results.ehm  -10429.47 20858.94 21176.94 21957.28 21473.53    159 0.306192       1

Result of model comparison

                    logLik deviance      AIC      BIC       HQ n_pars     chi p_value
results.mml2      -10447.69 20895.38 20975.38 21171.69 21049.99     40      NA      NA
results.davidian5 -10430.22 20860.45 20950.45 21171.29 21034.38     45 6.98715  0.2216
```

Normal latent trait: Data generation

- Sample size is 1000.
- 20 dichotomous items.
- Normally distributed latent trait.
- Item discrimination values range between 0.8 and 1.5.
- Item difficulties range between -2 and 2.



Normal latent trait: Parameter estimates

Item Discrimination

	Population	MML	KDE	EHM	LLS	Davidian1	Bayes
Item_1	1.0	0.8546848	0.8620040	0.8637855	0.8639600	0.8548655	0.8929545
Item_2	1.0	1.0136841	1.0270537	1.0443626	1.0307727	1.0139011	1.0158617
Item_3	1.0	0.9439529	0.9629743	0.9554326	0.9711763	0.9441548	0.9599756
Item_4	1.0	0.9796534	0.9858195	1.0025498	0.9849941	0.9798574	0.9883243
Item_5	1.5	1.4112043	1.4036278	1.4080727	1.3960961	1.4114805	1.3542426
Item_6	1.5	1.5982727	1.5896162	1.6039618	1.5796066	1.5986131	1.5209157
Item_7	1.5	1.5321684	1.5267631	1.5525320	1.5174316	1.5325076	1.4574698
Item_8	1.5	1.3236212	1.3186623	1.3378633	1.3109196	1.3238515	1.2830932
Item_9	0.8	0.8670351	0.8620659	0.8750655	0.8575204	0.8671369	0.8848905
Item_10	0.8	0.7528458	0.7486637	0.7490004	0.7478544	0.7529290	0.7871081
Item_11	0.8	0.9094363	0.9042325	0.9222566	0.8986800	0.9095418	0.9250804
Item_12	0.8	0.7048874	0.7012785	0.7123967	0.6991788	0.7049659	0.7446608
Item_13	1.0	0.8323680	0.8243179	0.8492721	0.8187892	0.8324462	0.8691298
Item_14	1.0	1.0149930	1.0032916	1.0160859	0.9971974	1.0150916	1.0210069
Item_15	1.0	1.0070163	0.9924263	1.0088243	0.9833178	1.0071170	1.0152081
Item_16	1.0	0.9371957	0.9273932	0.9467730	0.9194656	0.9372891	0.9584303
Item_17	1.5	1.5453277	1.5232622	1.5768282	1.5094580	1.5454177	1.4701269
Item_18	1.5	1.3231938	1.3039151	1.3314020	1.2955778	1.3232797	1.2909562
Item_19	1.5	1.5434005	1.5175095	1.5630303	1.5020868	1.5435050	1.4743252
Item_20	1.5	1.6124931	1.5833185	1.6335306	1.5667634	1.6125975	1.5330179

Item Difficulty

x	Population	MML	KDE	EHM	LLS	Davidian1	Bayes
Item_1	-2	-2.0393327	2.3713075	2.3607020	2.3672559	2.3858614	-2.3106797
Item_2	-2	-1.9602582	1.9184097	1.8860972	1.9147216	1.9336860	-1.9352417
Item_3	-2	-1.9817000	2.0705505	2.0740441	2.0585071	2.0992164	-2.0759669
Item_4	-2	-1.9004436	1.9336794	1.9012819	1.9359705	1.9398063	-1.9304900
Item_5	-1	-1.0238227	0.7368859	0.7319254	0.7435940	0.7256109	-0.7434676
Item_6	-1	-0.8889272	0.5683351	0.5647999	0.5751499	0.5563210	-0.5706806
Item_7	-1	-1.1878518	0.7869779	0.7772405	0.7946592	0.7753788	-0.7977565
Item_8	-1	-0.9356607	0.7169111	0.7074346	0.7233903	0.7070176	-0.7197277
Item_9	0	-0.0154580	0.0213173	0.0199702	0.0222381	0.0180423	-0.0173236
Item_10	0	0.0170300	-0.0199740	-0.0211200	-0.0193955	-0.0224058	0.0213889
Item_11	0	-0.0299846	0.0367862	0.0351727	0.0378991	0.0331833	-0.0344468
Item_12	0	0.0481377	-0.0661278	-0.0661981	-0.0658445	-0.0680733	0.0647067
Item_13	1	0.9436415	-1.1421248	-1.1139959	-1.1490816	-1.1333663	1.0979987
Item_14	1	1.0172750	-1.0107258	-0.9989744	-1.0166056	-1.0019305	0.9982224
Item_15	1	1.1009674	-1.1052541	-1.0893914	-1.1141790	-1.0929671	1.0899149
Item_16	1	0.9942735	-1.0692389	-1.0505907	-1.0773005	-1.0605816	1.0455323
Item_17	2	2.1671445	-1.4198675	-1.3848130	-1.4327504	-1.4020386	1.4449758
Item_18	2	1.9070130	-1.4588568	-1.4330035	-1.4689585	-1.4408700	1.4665407
Item_19	2	1.9296650	-1.2672146	-1.2402803	-1.2795732	-1.2499302	1.2861493
Item_20	2	2.0316618	-1.2780870	-1.2500474	-1.2909595	-1.2596113	1.2970855

Normal latent trait: Model fit

```
The best model: results.kde

                            HQ
results.mml2       20978.00
results.kde        20977.79
results.lls        20990.14
results.ehm        21425.23
results.davidian1  20981.87
```

Estimator	AIC	BIC
MML	20902.39	21099.70
KDE	20901.32	21102.54
LLS	20908.07	21124.01
EHM	21128.65	21908.98
Davidian1	20905.39	21106.61

Normal latent trait: Model fit

```
Result of model comparison

               logLik deviance      AIC      BIC       HQ n_pars      chi p_value
results.mml2 -10447.69 20895.38 20975.38 21171.69 21049.99     40       NA      NA
results.kde  -10431.73 20863.46 20945.46 21146.68 21021.94     41 31.91924       0

Result of model comparison

               logLik deviance      AIC      BIC       HQ n_pars      chi p_value
results.mml2 -10447.69 20895.38 20975.38 21171.69 21049.99     40       NA      NA
results.lls  -10429.82 20859.65 20947.65 21163.59 21029.72     44 8.933742  0.0628

Result of model comparison

               logLik deviance      AIC      BIC       HQ n_pars      chi p_value
results.mml2 -10447.69 20895.38 20975.38 21171.69 21049.99     40       NA      NA
results.ehm  -10429.47 20858.94 21176.94 21957.28 21473.53    159 0.306192       1

Result of model comparison

                    logLik deviance      AIC      BIC       HQ n_pars     chi p_value
results.mml2      -10447.69 20895.38 20975.38 21171.69 21049.99     40      NA      NA
results.davidian5 -10430.22 20860.45 20950.45 21171.29 21034.38     45 6.98715  0.2216
```

Zero-inflated latent trait

- Sample size is 1000.
- 20 dichotomous items.
- Normally distributed latent trait.
- Item discrimination values range between 0.8 and 1.5.
- Item difficulties range between -2 and 2.

Zero inflated latent trait: Parameter estimates

Zero-inflated

	a1	d	g	u
Item_1	2.059	-1.981	0	1
Item_2	2.074	-2.238	0	1
Item_3	1.685	-1.797	0	1
Item_4	1.947	-2.022	0	1
Item_5	1.853	-1.009	0	1
Item_6	2.055	-1.039	0	1
Item_7	2.043	-1.018	0	1
Item_8	2.190	-1.074	0	1
Item_9	2.183	-0.297	0	1
Item_10	1.969	-0.107	0	1
Item_11	1.942	-0.234	0	1
Item_12	2.078	-0.119	0	1
Item_13	2.171	0.884	0	1
Item_14	1.948	0.912	0	1
Item_15	2.156	0.971	0	1
Item_16	2.050	0.914	0	1
Item_17	2.389	1.993	0	1
Item_18	2.383	1.855	0	1
Item_19	2.488	2.206	0	1
Item_20	2.174	1.927	0	1

MML

	a1	d	g	u
Item_1	2.863	-4.563	0	1
Item_2	3.451	-5.199	0	1
Item_3	2.902	-4.427	0	1
Item_4	3.002	-4.975	0	1
Item_5	3.711	-3.999	0	1
Item_6	4.305	-4.839	0	1
Item_7	4.299	-4.682	0	1
Item_8	3.938	-4.397	0	1
Item_9	2.947	-2.685	0	1
Item_10	2.748	-2.569	0	1
Item_11	3.338	-2.982	0	1
Item_12	3.058	-2.839	0	1
Item_13	3.993	-2.355	0	1
Item_14	4.513	-2.663	0	1
Item_15	4.220	-2.452	0	1
Item_16	3.790	-2.151	0	1
Item_17	5.535	-2.477	0	1
Item_18	4.813	-2.123	0	1
Item_19	5.438	-2.473	0	1
Item_20	5.368	-2.435	0	1

Zero inflated latent trait: Parameter estimates

KDE

	a	b	c
Item_1	4.409177	1.3978533	0
Item_2	5.636680	1.3299835	0
Item_3	4.397736	1.3592409	0
Item_4	4.728037	1.4310231	0
Item_5	5.624791	1.0724763	0
Item_6	6.894050	1.0997855	0
Item_7	6.741461	1.0794417	0
Item_8	6.040299	1.0977017	0
Item_9	3.654917	0.9586850	0
Item_10	3.352186	0.9742456	0
Item_11	4.314778	0.9497904	0
Item_12	3.896256	0.9729607	0
Item_13	4.608335	0.7117920	0
Item_14	5.377529	0.7243417	0
Item_15	4.979571	0.7112516	0
Item_16	4.236694	0.6836507	0
Item_17	6.228922	0.6097777	0
Item_18	5.211947	0.5848797	0
Item_19	6.154123	0.6157416	0
Item_20	6.136058	0.6152857	0

LLS

	a	b	c
Item_1	8.991773	1.2227376	0
Item_2	12.025391	1.1855273	0
Item_3	8.985258	1.2037605	0
Item_4	9.700455	1.2378414	0
Item_5	11.304545	1.0655514	0
Item_6	14.085770	1.0795842	0
Item_7	13.429964	1.0703170	0
Item_8	12.169221	1.0784198	0
Item_9	6.113407	1.0067806	0
Item_10	5.205781	1.0160898	0
Item_11	7.718353	1.0027971	0
Item_12	6.905583	1.0147774	0
Item_13	7.598020	0.8640617	0
Item_14	9.669159	0.8799104	0
Item_15	8.451261	0.8673816	0
Item_16	6.324670	0.8342586	0
Item_17	10.399481	0.8089335	0
Item_18	8.294689	0.7850778	0
Item_19	10.322603	0.8128109	0
Item_20	10.600314	0.8154897	0

Zero inflated latent trait: Parameter estimates

EHM

	a	b	c
Item_1	1.923788	1.601666866	0
Item_2	3.110792	1.343297740	0
Item_3	2.000326	1.492636666	0
Item_4	2.115233	1.648337350	0
Item_5	2.551523	0.904151614	0
Item_6	3.171348	0.976229926	0
Item_7	2.825907	0.942053500	0
Item_8	2.663781	0.967768643	0
Item_9	1.474912	0.625902958	0
Item_10	1.363995	0.662275043	0
Item_11	1.627406	0.611525434	0
Item_12	1.567985	0.662288161	0
Item_13	1.781133	-0.006492926	0
Item_14	1.997166	0.010253299	0
Item_15	1.882681	-0.017521129	0
Item_16	1.631691	-0.074784612	0
Item_17	2.232520	-0.330023655	0
Item_18	2.026637	-0.347147978	0
Item_19	2.201882	-0.312625352	0
Item_20	2.254096	-0.305181493	0

Davidian 4

	a	b	c
Item_1	4.164533	1.2429116	0
Item_2	5.413031	1.1650260	0
Item_3	4.286210	1.1922690	0
Item_4	4.301161	1.2910377	0
Item_5	5.479770	0.8970190	0
Item_6	6.634449	0.9263638	0
Item_7	6.458365	0.9056654	0
Item_8	5.841126	0.9237167	0
Item_9	3.796553	0.7803134	0
Item_10	3.517286	0.7958987	0
Item_11	4.377569	0.7709448	0
Item_12	3.980643	0.7945476	0
Item_13	4.814541	0.5372792	0
Item_14	5.464859	0.5445770	0
Item_15	5.110710	0.5337078	0
Item_16	4.504708	0.5138290	0
Item_17	6.419464	0.4314044	0
Item_18	5.506167	0.4143847	0
Item_19	6.341067	0.4373846	0
Item_20	6.295866	0.4364065	0

Zero inflated latent trait: Parameter estimates

Bayesian item difficulty

1	1	-1.58
2	2	-1.52
3	3	-1.51
4	4	-1.65
5	5	-1.08
6	6	-1.14
7	7	-1.10
8	8	-1.12
9	9	-0.910
10	10	-0.931
11	11	-0.894
12	12	-0.927
13	13	-0.595
14	14	-0.592
15	15	-0.585
16	16	-0.573
17	17	-0.444
18	18	-0.443
19	19	-0.453
20	20	-0.451

Bayesian item discrimination

1	1	2.94
2	2	3.32
3	3	2.98
4	4	3.00
5	5	3.66
6	6	4.01
7	7	4.05
8	8	3.79
9	9	3.13
10	10	2.95
11	11	3.44
12	12	3.22
13	13	4.05
14	14	4.44
15	15	4.22
16	16	3.89
17	17	5.25
18	18	4.74
19	19	5.17
20	20	5.13

Summary

- When the latent trait does not follow a normal distribution, the model parameter estimates based on the standard MML can be deleteriously impacted.
- There exist several alternative estimators that have been shown to be useful when the latent trait is not normally distributed (e.g.,Lee & Li, 2026; Woods, 2015.
- These approaches can be carried out in the R software environment.
- The examples demonstrated how these methods can be applied to various modeling problems.
- For a zero-inflated latent trait, none of the alternatives except for the zero-inflated IRT model produced reasonable estimates of the item parameters.